

Design and Implementation of a Database Management System for Municipal Waste Collection and Recycling Management System

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Problem statement

Manual waste management breaks down:

- Pickups get missed with no record of why
- Vehicles or crews get double-booked on overlapping routes
- Recycling output isn't tracked against input, hiding losses
- Household complaints go unresolved with no accountability trail

Our solution

A relational database that automates ward and route planning, vehicle/crew assignment, pickup logging, recycling batch tracking at transfer stations, and complaint resolution — end to end.

Objectives

1

No overlapping assignments

A vehicle or crew can't be scheduled on two routes at once

2

Non-negative weights

All pickup and recycling weights must be zero or positive

3

Pickups on valid schedules

Every pickup log must tie back to a real, existing schedule

4

Recycling output within input

A batch can't recycle more than what came in

5

Closure details required

Every recycling batch must record how it was closed out

6

Operational reporting

Missed pickups, route performance, recycling rate, complaints

Users & scope

Who interacts with the system, and what it covers

Admin / Municipal Officer

Registers wards, points, plans routes and schedules, assigns vehicles and crews

Collection Crew

Executes pickups on assigned routes, logs weight and status per point

Household / Resident

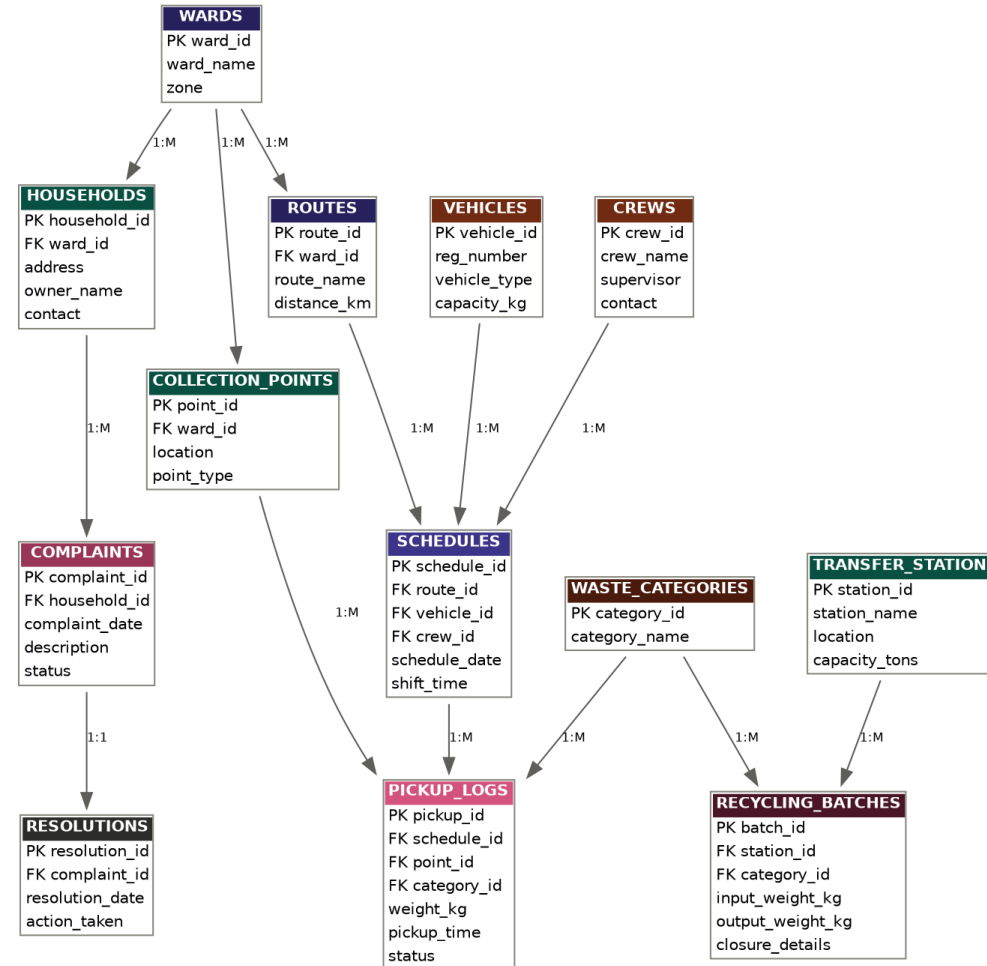
Source of pickups; raises complaints when service is missed

Scope

Ward & collection point registration · route & schedule planning · vehicle/crew assignment · pickup logging · recycling batch tracking at transfer stations · complaint handling · reporting

Entity-relationship diagram

13 entities modelling the full waste-collection lifecycle



Key relationships

1 Ward (1) → Households (M) | Ward (1) → Collection Points (M) | Ward (1) → Routes (M)

2 Route (1) → Schedules (M)

3 Vehicle (1) → Schedules (M) | Crew (1) → Schedules (M)

4 Schedule (1) → Pickup Logs (M) | Collection Point (1) → Pickup Logs (M)

5 Waste Category (1) → Pickup Logs (M) | Waste Category (1) → Recycling Batches (M)

6 Transfer Station (1) → Recycling Batches (M)

7 Household (1) → Complaints (M)

8 Complaint (1) → Resolution (0 or 1)

Initial relational schema

schedules	pickup_logs	recycling_batches	complaints
schedule_id PK	pickup_id PK	batch_id PK	complaint_id PK
route_id, vehicle_id, crew_id FK	schedule_id, point_id, category_id FK	station_id, category_id FK	household_id FK, status
Rule: no overlapping vehicle/crew slots	Rule: CHECK weight_kg >= 0	Rule: CHECK output_weight <= input_weight	Rule: status: open / resolved

Business rules → database constraints

Non-negative weights

```
CHECK (weight_kg >= 0)
```

Recycling output within input

```
CHECK (output_weight_kg <= input_weight_kg)
```

No overlapping vehicle/crew assignment

```
UNIQUE (vehicle_id, schedule_date, shift_time)
```

Closure details required

```
closure_details set to NOT NULL
```